Statistics

Definition

Statistics is the branch of mathematics that deals with collecting, organizing, analyzing, interpreting, and presenting data.

statistics helps us understand data and make better decisions.

* Importance of statistics

1. It helps in collecting and organizing data.
2. It helps to analyze large amounts of information.
3. It helps to compare different groups.
4. It helps to identify patterns and trends.
5. It helps in decision-making.
6. It is widely used in business, education, healthcare, and machine learning.

Types of Statistics

There are two types of statistics:

A) Descriptive statistics.

→ Descriptive statistics summarizes and describes the main features of a dataset.

Examples:

- Mean
- Median
- Mode
- Range
- variance
- Standard Deviation
- Charts and Graphs

B. Inferential Statistics

→ Inferential statistics uses sample data to draw conclusion about a larger population

Example

- Hypothesis Testing
- Confidence Intervals
- Estimation
- Prediction

Important Terms

1. Population: The complete group of individuals or observations being studied.

2. Sample: A smaller part of the population selected for study.

3. Data: facts, figures, or information collected for analysis.

4. Variable: A characteristics that can take different values.

5. Observation: A single recorded value or data point.

Central Tendency

Central tendency describes the center or typical value of a dataset.

The three main measures of central tendency are

1. Mean
2. Median
3. Mode

Applications of Statistics

- Education : Analyzing student marks.
- Business : Comparing sales and profits.
- Healthcare : Studying diseases and treatments.
- Finance : Analyzing income and investme

* Conclusion : Statistics is an important tools for understanding data, identifying patterns, and making informed decisions.

Mean, Median and Mode

Measures of central tendency

→ Measures of central tendency identify the central or typical value of a dataset.

The three main measures are Mean, Median, and Mode.

A. Mean (Average)

The mean is calculated by dividing the sum of all observations by the total number of observations.

Formula:

Mean = Sum of all values / Number of values

Example:

Data: 60, 70, 80, 90

Sum = 60 + 70 + 80 + 70 + 90 = 360

Number of values = 5

Mean = 360/5

Mean = 72

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5. Median

The median is the middle value of a dataset when the values are arranged in ascending or descending order.

When the number of observations is odd:

Median position $= (n+1)/2$

Example:

Data : 2, 4, 6, 8, 10

$n = 5$

Median position $= (5+1)/2 = 3$rd position

Median $= 6$

When the number of observations is even:

Median = (value at $n/2$ position + value at $(n/2+1)$ position)/2.

Example :

Data: 2, 4, 6, 8

Median = (4+6)/2

Median = 5

C. MODE

The mode is the value that occurs most frequently in a dataset.

Example :

Data : 2, 3, 3, 4, 5, 3, 6

The value 3 occurs three times.

Mode = 3

A dataset can have :

- one mode (unimodal)
- two modes (bimodal)
- More than two modes (multimodal)